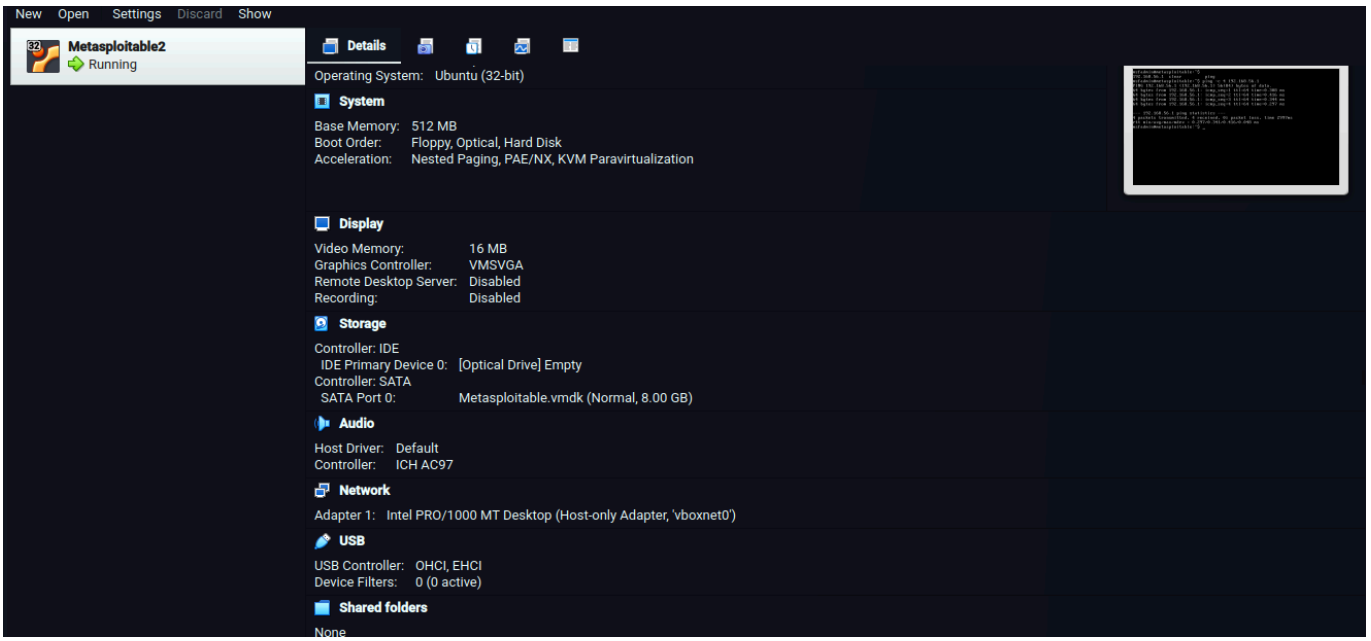
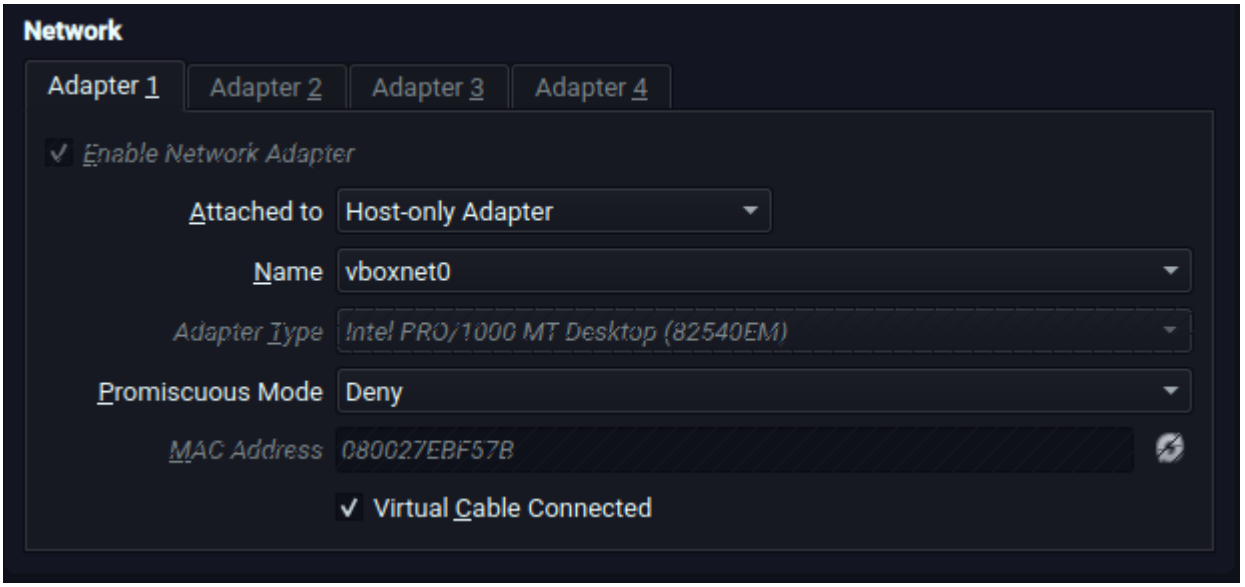
Assignment 3

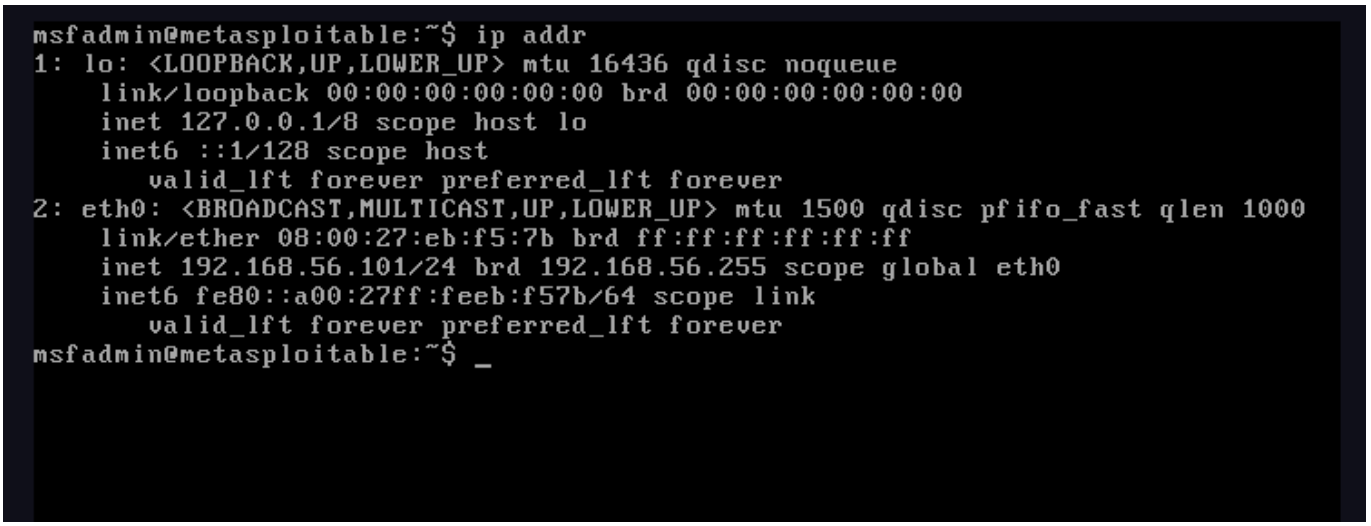
```
Name : Ravish Ranjan
Roll no. : 45
Course : MCA
Subject : Information Security
```

Part A

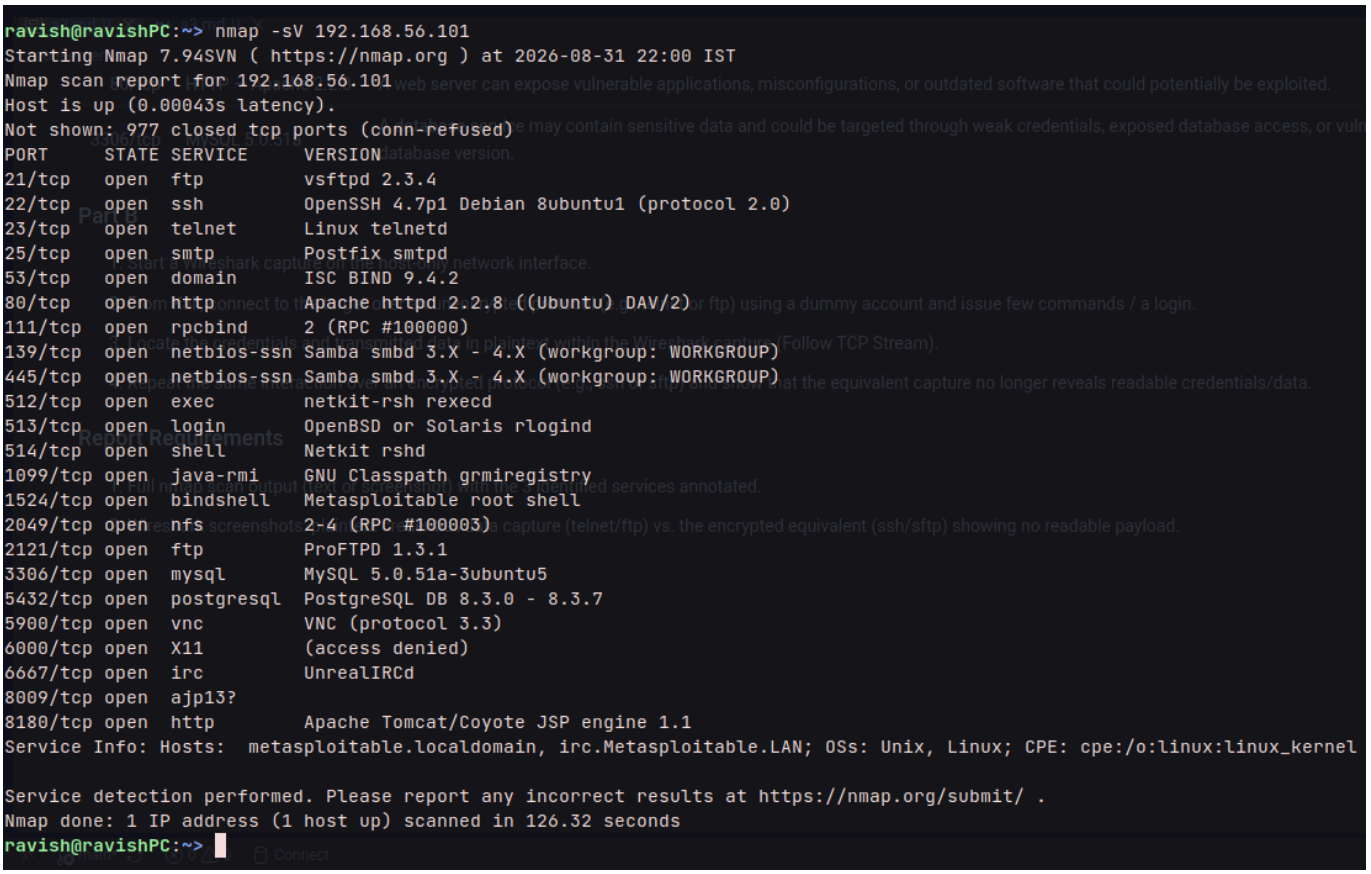
1. Set up a second VM (e.g. Metasploitable2) on the same host-only/internal VirtualBox network as your Kali VM.







2. From Kali, run nmap scans against the target: a basic scan, then nmap -sV -A for service/version and OS detection.



3. Identify at least 3 open ports/services and, for each, state one reason an intruder might target it.

Port	Service / Version	Why an intruder might target it
21/tcp	FTP — vsftpd 2.3.4	FTP may allow weak/anonymous authentication or expose files. The identified version is also an old version which may have security issues.
22/tcp	SSH — OpenSSH 4.7p1	SSH provides remote login access. An attacker could target weak credentials or vulnerabilities in an outdated SSH implementation.

Port	Service / Version	Why an intruder might target it
80/tcp	HTTP — Apache 2.2.8	A web server can expose vulnerable applications, misconfigurations, or outdated software that could potentially be exploited.
3306/tcp	MySQL 5.0.51a	A database service may contain sensitive data and could be targeted through weak credentials, exposed database access, or vulnerabilities in the old database version.

Part B

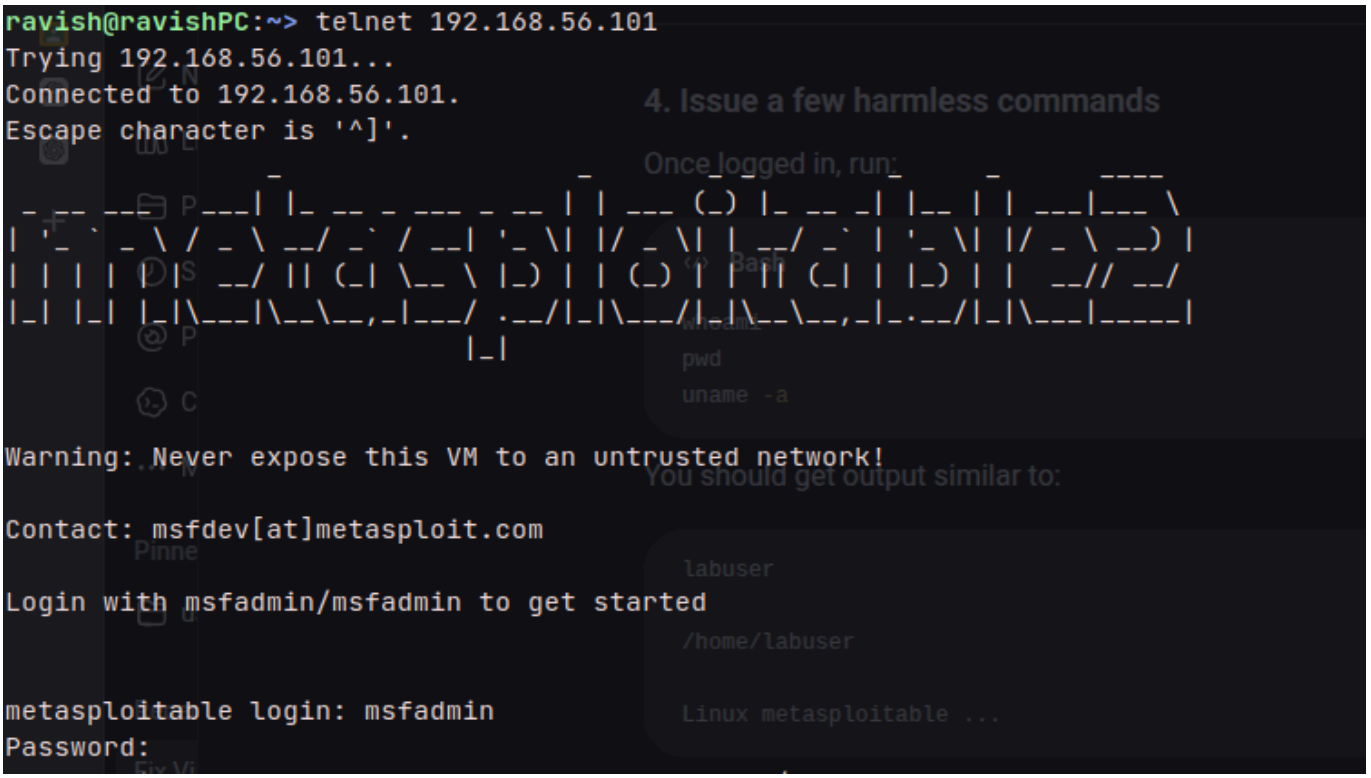
1. Start a Wireshark capture on the host-only network interface.

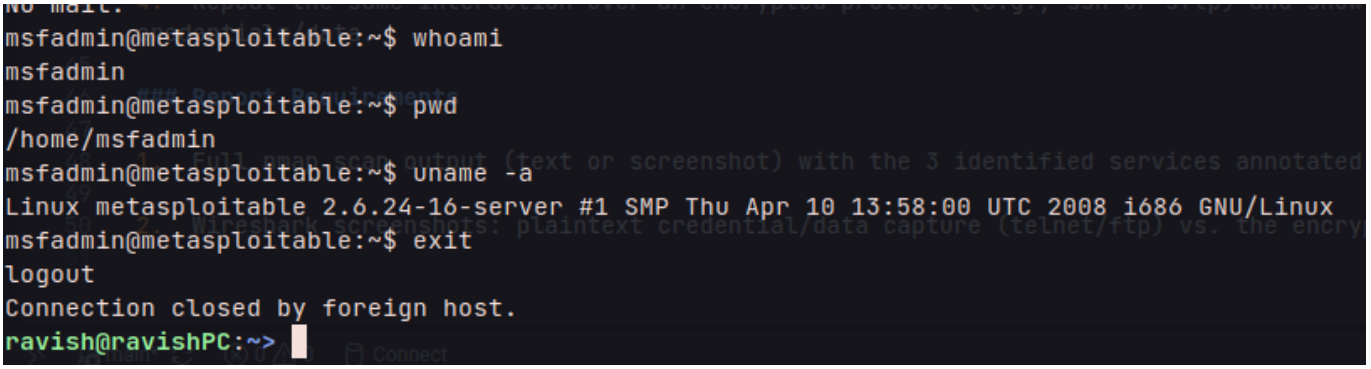
```
ravish@ravishPC:~> ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.662 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.319 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.343 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.318 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3083ms
rtt min/avg/max/mdev = 0.318/0.410/0.662/0.145 ms
ravish@ravishPC:~>
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	0a:00:27:00:00:00	Broadcast	ARP	42	Who has 192.168.56.101? Tell 192.168.56.1
2	0.000379443	PCSSystemtec_eb:f5:…	0a:00:27:00:00:00	ARP	42	192.168.56.101 is at 08:00:27:eb:f5:7b
3	0.000396743	192.168.56.1	192.168.56.101	ICMP	98	Echo (ping) request id=0xb930, seq=1/256, ttl=64 (reply in 4)
4	0.000621405	192.168.56.101	192.168.56.1	ICMP	98	Echo (ping) reply id=0xb930, seq=1/256, ttl=64 (request in 3)
5	1.035148548	192.168.56.1	192.168.56.101	ICMP	98	Echo (ping) request id=0xb930, seq=2/512, ttl=64 (reply in 6)
6	1.035442719	192.168.56.101	192.168.56.1	ICMP	98	Echo (ping) reply id=0xb930, seq=2/512, ttl=64 (request in 5)
7	2.035125442	192.168.56.1	192.168.56.101	ICMP	98	Echo (ping) request id=0xb930, seq=3/768, ttl=64 (reply in 8)
8	2.035431493	192.168.56.101	192.168.56.1	ICMP	98	Echo (ping) reply id=0xb930, seq=3/768, ttl=64 (request in 7)
9	3.083139486	192.168.56.1	192.168.56.101	ICMP	98	Echo (ping) request id=0xb930, seq=4/1024, ttl=64 (reply in 10)
10	3.083427627	192.168.56.101	192.168.56.1	ICMP	98	Echo (ping) reply id=0xb930, seq=4/1024, ttl=64 (request in 9)
11	4.990452918	PCSSystemtec_eb:f5:…	0a:00:27:00:00:00	ARP	42	Who has 192.168.56.1? Tell 192.168.56.101
12	4.990468518	0a:00:27:00:00:00	PCSSystemtec_eb:f5:…	ARP	42	192.168.56.1 is at 0a:00:27:00:00:00

2. From Kali, connect to the target over an unencrypted protocol (e.g., telnet or ftp) using a dummy account and issue few commands / a login.

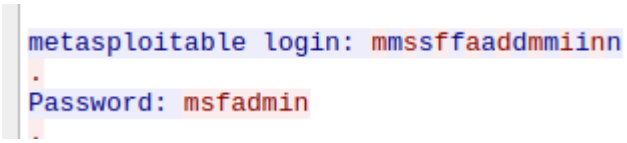




3. Locate the credentials and transmitted data in plaintext within the Wireshark capture (Follow TCP Stream).

Unencrupted Data stream





4. Repeat the same interaction over an encrypted protocol (e.g., ssh or sftp) and show that the equivalent capture no longer reveals readable credentials/data.

Encrupted Data stream

```
SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.18
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
.........#....1.+..X.....1sntrup761x25519-sha512@openssh.com,curve25519-sha256,curve25519-sha256@libssh.org,ec
dh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,diffie-hellman-gro
up16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,ext-info-c,kex-strict-c-v00@openssh.co
m....ssh-ed25519-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@openssh.com,ecdsa-sha2-nistp384-cert-v01@op
enssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.com,sk-ssh-ed25519-cert-v01@openssh.com,sk-ecdsa-sha2-nistp256-
cert-v01@openssh.com,rsa-sha2-512-cert-v01@openssh.com,rsa-sha2-256-cert-v01@openssh.com,ssh-ed25519,ecdsa-sha
2-nistp256,ecdsa-sha2-nistp384,ecdsa-sha2-nistp521,sk-ssh-ed25519@openssh.com,sk-ecdsa-sha2-nistp256@openssh.c
om,rsa-sha2-512,rsa-sha2-256,ssh-rsa...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-
gcm@openssh.com,aes256-gcm@openssh.com...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes12
8-gcm@openssh.com,aes256-gcm@openssh.com....umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm
@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,
hmac-sha2-256,hmac-sha2-512,hmac-sha1....umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@op
enssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hma
c-sha2-256,hmac-sha2-512,hmac-sha1....none,zlib@openssh.com,zlib....none,zlib@openssh.com,zlib..............
........
.k...l,.s.7.........~diffie-hellman-group-exchange-sha256,diffie-hellman-group-exchange-sha1,diffie-hellman-gr
oup14-sha1,diffie-hellman-group1-sha1....ssh-rsa,ssh-dss....aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,arcfo
ur128,arcfour256,arcfour,aes192-cbc,aes256-cbc,rijndael-cbc@lysator.liu.se,aes128-ctr,aes192-ctr,aes256-ctr...
.aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,arcfour128,arcfour256,arcfour,aes192-cbc,aes256-cbc,rijndael-cbc
@lysator.liu.se,aes128-ctr,aes192-ctr,aes256-ctr...ihmac-md5,hmac-sha1,umac-64@openssh.com,hmac-ripemd160,hmac
-ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96...ihmac-md5,hmac-sha1,umac-64@openssh.com,hmac-ripemd160,hmac-
ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96....none,zlib@openssh.com....none,zlib@openssh.com.............
.............".......... .....................e .......kN..^....!..x.......5......}...R..   X.Z.....l..l..P.P.
...B..)..*.IX...8..!........;.<LZ.k.6.D......rw.6.q..
}.#T....{7mr4QG........w..o..iZ......O.....:qP.Y&#..+'u.|...E..u...t.tr.5}.C.....A..K.L@H...k...<....3j..)...N
..w.rdlb.7.e..".q.|..=....g.H..".{..a....%...m...Y.yJ...N...."v..f..
..Q.|G.....e.......}...u>*@0..{.B..B8E...#..f1.     ..<`1/.G..@=g..ve......0QM.}~...|..xl..................  ..
..c........+.h.......K,.2w..7}H..v..s
m...W.&...1P...iG@.w..4x[p..........s.>...H.2.....N&.o._K.j.0uk..z..h.:.mD(v.d.+...V..v...D...x. ..O.e...c.p.
..x..Zw............
...S
b..Cc..x.9......./.....1.I..-../0...D.[2..#.shY.tf.....].C......U..v=...].=...0>.8..-.=..`....X.,wf...^.$~....
..2z.....{R.>)C..-.x%'..l.C*Q..Ygh..P.f..=...#vT..$...?L.(.'...{.=..."b.......-.......&.%...;q."...@.........
...
|       ssh-rsa     #          S.9    d     O  ZB.5    F..8.x      #     p..P      <O.`09..F.JT       X     FR.?    b
```

```
!........ssh-rsa....#.........S.9...d....Q..ZB.5...F..8.x.....#....p..P.....<O.`09..F.JT......X....FR.?...b...
H..s...{   ."'}..{v.m...B\V..'..(..5.@).....s....9r;.0..T.....#...q...........;..Da~.O..]....ogz.$..V.$D..2..5
+.....62ga9......w....6,..9_.q....@q..Q..~Bk.-.].o..t.S.......{....\%.......S>8.......0....M.t/O.      ..FJ.y?g
\...ym.;.1.<......!.l...9g..<y$X....Ow.#..&........FVQ;..j..mj.........i.zZ5FCS.=A.xQ.Z^     TmK>.....d.%..q..!
...#....:.<M...d...,k.S..&.yi,..Z.J....U.pe...^|....NpY.....V..17.v._. .......T......6..9...H.kkD...M=,.Fc.{
          |4.1..._.F.D..qZ.n.  ).}..S..-.KQ..,.D*..Mrn5wz..>~.SQ3.f.Zp.O7.?#Yk.!.'|..S...5M.).:...$N>6j&..t'.d.
.x../....N...$.`....dH.i..Ua;........ssh-rsa.....k....P.QHY....|.Q;N..E.a..Iyp..\..._.....kX.........{f?..e.g.
`..=^.s`.v.Z.#.._s....e#......{4          .`$/...Et.OG..;p.~H..?.JOy....~y.........NOo.R.6..)..~...B..!:/p.....
.a......+58I.B.(nL..
O
.b8M....@..-{.!..n.x........A}-.'.-.l..K(.).M.l..)'[."..
.vR.....EO..............
...........
...........T......)6SP}K..\...2X0./.....l.&/..$?K..Q...........:.-2...._.n...e`q.k#.nr....2F
I.....!.......>.#...FS.C~....\...1.}..4...O.}N.&l.S......._J.H..W".r.:.C.HZ...r.,.rl.c.Y...a....|....6...V\...
.`..&.....~..Y-........v)..&{..uE..n.p    I.F..?..w(n.o.m t....l.Bj..*i.OH.*..u...0o.}...J....4....7..b.s...i.+
...}...D@...)%?r..!(j[)...{!._.9.<....j...q5..r.#..Q.....P....hK.nH...R.H.2.#........uaI.%!A......u.N1......t.
.-..M..PsX.p..@.{65g.J...<?.f.>.O_d.0...G..SQ.8I..+.........r...)U.P.A.<           ..8.g...6.y.a..i../..<.=..[.
I..4.g..O.`....c.k.......*..........!*!3..........,=.nc.,2.}..3..|.........-tR.e.M.....fO.S.F.....j......~i..
t..M.(y.=<.....S....N....z:V.EI}.&(...L.....o..u..T.....T..1...0.......<.J..,.?...e D.;l... ~F.z..=..k.-.v.zC.
%..#.-)D=.R.nd.l;.7?.;:..mu#(.SJW....H....-.'....?...:.h<..a..U.
..f...+...A.z..m........`f:.:....U.....(@2.....H.... ._.V..A/KH.-r...[..q...$}...........g..(2]..j]k@..r.I.p.
phi..O..3...FV.F
{q"....?r..d..EL,.`P.4.....?../.~.....m.~=..+@.].q.mk.!1...5...!3.y....~v.DP.4Vt~`..[ORo`.u.d'T....9h.C.t..@.,
.0..^..nIs._j..%.G2..W....E........;n...G29wT5.k.f.RE..\...0H(.........
E8L.C....K..i.....
..../...Y(}.t..n.q.m.k..E.....!..+,...9.=g.V.O%.p.s.k.=.:%.`...vB...5..r
....N...S.'..tb.     .Y..FguK.m..C..+.I
_......p8.........Y.........:....mo.w.....6..
.Q:m.K.R.Fx4E3B..KM.H....Z..{.............H...I....t...W.6_,..C..5..mKm_3?.e.o...H..[.lTuY..b.]...hb.( ...q]<.
.n.S.......8<.3.........u.na4%...u...
..a!&..`F83.....c.~.E.2.x..mN.z.Y..%&K..m..+.ots.\K;3..       x{.d.h..b..R........^V..Gh..."..-...dc~B.\.......
...e..?gv.)B.Rk9.<`.....`3..T...e,._+&?..t.%...,..G3.~.].a9..Z:l..........v.D..e.z..L#-.C....I@.K|....gF...jF.
.;..O...$>k88....%e.&.7aH.9K8..{.<..;..).....{...lQ.#..  .R.^7..u..       V.?q$..    .)_W...u.!e3g...\...uP...*r
......LW....".!V...5.m!.
.G.>...W..PNL...>|S....#.8...^-...i..2.[T.....V...=.          .t)].J....!....Ae...`........B..\....9.c.a...l.^
....&....J.;.`.&..|.&.Q.s.&tB.H..h.i ..o......q.4...j.U...Sbff....... .L.I.....u2.+..
```

